

Q1.

```
package Test_21_01_26;

import java.util.Scanner;

public class Q1 {

    public static long fact(int n)
    {
        if(n == 0 || n == 1)
        {
            return 1;
        }

        return n * fact(n - 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number : ");
        int n = sc.nextInt();

        System.out.println("Factorial of " + n + " is : " + fact(n));
    }
}
```

Q2.

```
package Test_21_01_26;

import java.util.Scanner;

public class Q2 {

    public static void main(String[] args) {

        int a = 0, b = 1, c;
```

```

        for(int i=1; i<=3; i++)
        {
            for(int j=1; j<=i; j++)
            {
                c = a + b;
                System.out.print(c + " ");

                a = b;
                b = c;
            }

            System.out.println();
        }
    }
}

```

Q3.

```

package Test_21_01_26;

public class Q3 {

    public static void main(String[] args) {

        int n = 3;
        int i, s, j, r;

        // Pattern 1

        for(i=n; i>=1; i--)
        {
            for(s=1; s<i; s++)
                System.out.print(" ");

            for(j=n; j>=i; j--)
                System.out.print(j);

            for(r=j+2; r<=n; r++)
                System.out.print(r);

```

```

        System.out.println();
    }

    for(i=2; i<=n; i++)
    {
        for(s=n-1; s<=i; s++)
            System.out.print(" ");

        for(j=n; j>=i; j--)
            System.out.print(j);

        for(r=j+2; r<=n; r++)
            System.out.print(r);

        System.out.println();
    }

    System.out.println("");

    // Patters 2
    for(i=1; i<=3; i++)
    {
        for(j=3; j>=i; j--)
        {
            System.out.print("* ");
        }
        System.out.println();
    }

}

}

```

Q4.